Step 1 – Install Required Software

Install the following software before running the project:

- Windows 10 / 11 (64-bit)
- Python 3.8 to 3.10 (Recommended for TensorFlow compatibility)
- pip package manager
- Integrated Webcam or USB Camera (for leaf capture)

Download Python from:

https://www.python.org/downloads/

Verify Python installation:

python --version

Step 2 – Locate Project Files

Ensure the project folder is extracted and placed exactly at:

D:\project\

Open Command Prompt or PowerShell and move to the project folder:

cd D:\project

Step 3 – Create and Activate Virtual Environment

Create virtual environment:

python -m venv .venv

Activate virtual environment:

.\venv\Scripts\activate

Step 4 – Install Required Libraries

Install all required Python libraries needed for the AI and Interface:

pip install tensorflow tensorflow-hub opencv-python numpy PyQt5 scikit-image

Step 5 – Start the Application

Open your Terminal, ensure your virtual environment is activated, and run:

cd D:\project\projectf

.\venv\Scripts\python.exe prfinal.py

Step 6 – Run Detection & View Results

Once the Plant Disease Detection dashboard opens:

1. Login / Register: Enter your credentials to access the secure system.

2. Real-Time Detection: Place a leaf in front of the camera or upload an image to predict the disease.

3. Compare Leaves: Navigate to the comparison tool, upload a Past Scan and a Present Scan to track if the disease severity has increased or decreased.

4. View Diagnostics: Review the Infection Severity %, AI Focus (Grad-CAM), and Treatment Recommendations directly on the dashboard.

5. History Logs: Check the System Diagnostic Logs to track and manage previously saved reports.